

Algebra I Proportional Relationships Cheatsheet

Module 4

Ratios and Rates

A ratio compares two quantities.

$$a : b, \quad \frac{a}{b}, \quad a \text{ to } b$$

A unit rate has denominator 1.

$$\frac{\text{miles}}{\text{hours}} = \text{miles per hour}$$

Proportions

$$\frac{a}{b} = \frac{c}{d} \quad (b \neq 0, d \neq 0)$$

Cross products are equal:

$$ad = bc$$

Direct Variation

$$y = kx$$

The constant of proportionality is

$$k = \frac{y}{x}$$

Graph feature:

$$y = kx \text{ passes through } (0, 0)$$

Slope as Rate of Change

$$m = \frac{\text{change in } y}{\text{change in } x} = \frac{y_2 - y_1}{x_2 - x_1}$$

For proportional relationships:

$$m = k$$

Percent Change

$$\text{percent change} = \frac{\text{new} - \text{old}}{\text{old}} \cdot 100\%$$

Increase:

$$\text{new} = \text{old}(1 + r)$$

Decrease:

$$\text{new} = \text{old}(1 - r)$$

where r is written as a decimal.

Tables and Graphs

A relationship is proportional when:

- the ratio $\frac{y}{x}$ is constant for all nonzero x ,
- the graph is a line,
- the line passes through the origin.

Quick Checks

- Keep units attached to rates.
- Proportional means linear through the origin.
- A constant difference is linear; a constant ratio through the origin is proportional.
- Convert percents to decimals before multiplying.